- **3-Stages of Problem Identification**

→Domain : Machine Learning

- Problem : Predict Chronic Kidney Disease (CKD)

→Supervised Learning Model : ML based on labeled data

- Input → patient medical details
- Output → CKD / Not CKD

→Model Type **:** Classification Model

- Since the output is categorical

- **The Finale Machine Learning best method of Classification**

  - Applied multiple classification algorithms on the dataset
  - Algorithms used include Logistic Regression, Naive Bayes, and KNN
  - Models were trained and evaluated using standard metrics
  - Among all models, Logistic Regression performed better
  - It provided more accurate and consistent results
  - Hence, Logistic Regression was selected as the final model

- **Confusion Matrix - Logistic Regression**

```
print(clf_report)
```

```
              precision    recall  f1-score   support

       False       0.98      1.00      0.99        45
        True       1.00      0.99      0.99        75

    accuracy                           0.99       120
   macro avg       0.99      0.99      0.99       120
weighted avg       0.99      0.99      0.99       120
```

➢ **Accuracy (0.99)**

→ Overall model prediction is 99% correct

→ Indicates very high performance

- **Class: False (No Disease)**
  → Precision: 0.98 → Few false predictions
  → Recall: 1.00 → No actual cases missed
  → F1-score: 0.99 → Good balance
- **Class: True (Disease)**
  → Precision: 1.00 → All predicted are correct
  → Recall: 0.99 → Very few missed cases
  → F1-score: 0.99 → Strong performance
- **Macro Average (0.99)**
  → Average performance of all classes equally
  → Model works balanced for both classes
- **Weighted Average (0.99)**
  → Average based on data distribution
  → Model performs well even with different class sizes
- **Confusion Matrix**
  → True Positive (TP): Correct disease prediction
  → True Negative (TN): Correct no-disease prediction
  → False Positive (FP): Wrong disease prediction
  → False Negative (FN): Missed disease case
- **Overall Result**
  → High TP and TN values
  → Very low FP and FN
  → Model has low error and high accuracy